

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Unit Test 4

Unit Test 4 - Answer Key

Class: Class 10

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL Database Management

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** b - ORDER BY

- **Explanation:** ORDER BY sorts the result set in ascending or descending order. There is no SORT BY or ARRANGE BY in SQL.

2. **Answer:** c - CREATE

- **Explanation:** CREATE is a DDL (Data Definition Language) command used to create database objects. SELECT, INSERT, UPDATE are DML commands.

3. **Answer:** a - SELECT - FROM - WHERE - GROUP BY - ORDER BY

- **Explanation:** The correct SQL clause order is: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY.

4. **Answer:** b - COUNT()

- **Explanation:** COUNT() returns the number of rows. SUM() returns total, there is no TOTAL() or ROWS() function.

5. **Answer:** b - UNIQUE

- **Explanation:** UNIQUE constraint ensures all values in a column are different. PRIMARY KEY also ensures uniqueness but cannot contain NULLs.

Section II: Short Answer (5 x 1 = 5 marks)

1. **SQL Queries for Products table**

- **Answer:**

-- Products with price between 100 and 500

```
SELECT * FROM Products WHERE price BETWEEN 100 AND 500;
```

-- Count products in each category

```
SELECT category, COUNT(*) FROM Products GROUP BY category;
```

-- Product with maximum price

```
SELECT * FROM Products WHERE price = (SELECT MAX(price) FROM Products);
```

-- Products sorted by stock descending

```
SELECT * FROM Products ORDER BY stock DESC;
```

- **Marking:** 0.5 marks for each correct query.

2. **Foreign key**

- **Answer:** A foreign key is a column that references the primary key of another table. It establishes a parent-child relationship. Example: Orders.customer_id references Customers.customer_id. This ensures referential integrity - you cannot add an order for a non-existent customer.
- **Marking:** 1 mark for definition, 1 mark for example and relationship.

3. SQL Output

- **Answer:**

```
SELECT ROUND(45.678, 2);      -- 45.68
SELECT MOD(17, 5);           -- 2
SELECT NOW();                 -- Current date and time (e.g., 2024-01-15 10:30:00)
```

- **Marking:** 0.5 marks for each correct output.

4. DELETE vs DROP

- **Answer:** DELETE removes rows from a table (DML command, can be rolled back with WHERE clause). DROP removes the entire table including structure (DDL command, cannot be rolled back). DELETE: DELETE FROM Students WHERE roll=1. DROP: DROP TABLE Students.
- **Marking:** 1 mark for each difference.

5. JOIN with example

- **Answer:** JOIN combines rows from two tables based on a related column. INNER JOIN returns only matching rows.

```
SELECT Students.name, Marks.marks
FROM Students
INNER JOIN Marks ON Students.roll = Marks.roll;
```

- **Marking:** 1 mark for JOIN explanation, 1 mark for correct query.